

A NOTE ON DETERMINANTS OF HOUSEHOLD SAVING BEHAVIOR

Alf LINDQVIST *

The Stockholm School of Economics, Sweden

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By using cross-sectional interview data from 429 randomly selected households, four different saving estimates, namely reported bank savings, repayments of debts, total savings and a liquidity estimate, are computed and analyzed separately. Both socioeconomic characteristics and “softer” variables, such as attitudes and expectations, are used as explanatory variables.

In a regression analysis the predictors used, fail to explain bank savings, while the traditional socioeconomic variables are found to influence the debt measure. The behavioral predictors are more successful in explaining the liquidity measure. When the regression result with the liquidity measure as the dependent variable (with the highest R^2 value) is decomposed using a hypothetical causal model and path analysis, it appears that the effects of household income and educational level of the family head are substantial. However, the intervening behavioral variables are still significant.

Introduction

How households allocate their income on different types of saving and consumption affects economic development in the short as well as in the long run. It affects demand and also the capacity to invest—and thereby price level, business cycles and growth rate.

However, household's saving behavior does not only entail economic consequences on the macro level. From the point of view of society, saving is often considered to be connected with highly valued individual attributes such as industriousness and ability to plan ahead. A decrease in people's interest in saving may indicate that parts of the population are changing their life style toward a more materialistic way of living.

The allocation of household income on saving and consumption, and

* Author's address: The Economic Research Institute, The Stockholm School of Economics, Box 6501, 11383 Stockholm, Sweden.

the change in allocation from one period to another on different types of consumer spending and saving, may serve as indicators of the varying degree to which monetary and socio-political objectives are reached. Economic political objectives such as a decrease in purchasing power (to affect inflation rate and terms of trade) or economic improvement for poor population groups are all directly associated with income use.

Both with economic and non-economic objectives in mind, the ability to predict the development of household savings in the long as well as in the short run and the ability to influence household saving are thus of great importance.

Saving has mostly been studied by economists concerned with the effects on spending and saving of such environmentally conditioned factors as rates of return on financial assets and level of income and wealth (Crockett 1977). The models of household behavior have implications for aggregate consumption and saving, and attention has been devoted to quantifying the aggregate relationships. However, little attention has been given to decision processes or to variables within the individuals or households to increase the predictive value of the models. Ölander and Seipel (1970) state three causes for this. Firstly, economics is usually treated as a deductive science, which deduces principles of economic behavior from features of human nature assumed to be valid at all times. Secondly, the main emphasis is on studying aggregate behavior. Individual differences in saving are assumed to depend on never-changing differences in taste and on random fluctuations, tending to cancel out when data are aggregated. Thirdly, most of the attempts to provide realistic descriptions of economic behavior have been devoted to buyer behavior. Saving is often considered as a residual (*cf.* Katona 1980).

The most comprehensive studies focusing on the predictive value of psychological determinants of economic activities have been carried out by Katona and his colleagues. The results obtained have sometimes differed greatly from traditional economic insights and have contributed to the understanding of economic development.

An extensive and critical review of the literature and the empirical findings from different behavioral sciences which aim at the same goal of explaining individual saving is given in Ölander and Seipel (1970).

The project

The research project which is partially reported here is an attempt to make predictions of savings. A micro approach is applied where (1) households are studied and (2) other indicators besides the traditional economic ones are anticipated to influence saving behavior.

As a first step in the attempt to assess causal factors influencing saving, it is necessary to obtain measurements involving registration of household savings and changes in household savings. It is both difficult and expensive to gain precise knowledge of how much different households save, or when changes occur. In this project simple and inexpensive instruments are desirable, since they are aimed to be applicable for regularly recurring studies on representative samples of the population. For this reason it is important to develop appropriate measuring instruments, whose primary objective is not to measure exact amounts, but rather to indicate scope and change.

The aims of the project were thus:

- (1) to develop instruments yielding indicators of the amount of household saving;
- (2) to develop and test short- and long-run predictors of household's saving;
- (3) to try out these instruments in representative studies of the Swedish population.

Three studies have been carried out so far, and they have primarily dealt with methodological problems. Three reports of the project, one for each study, have hitherto been published (Julander *et al.* 1977; Lindqvist *et al.* 1978; Lindqvist 1980).

This paper

This paper reports on a specific part of the results from the third and most comprehensive study.

Methodological issues have been the main emphasis (aim 1), but the study has generated data concerning household saving behavior. Analyses of the influence of an *a priori* set of predictors upon saving are presented and discussed in the paper (aim 2).

The measurements and statistical techniques used, partly in conjunction with causal theory, involve a number of methodological problems and assumptions. A discussion of these problems is, however, omitted. For a fuller treatment see some of the standard texts, for example: Blalock (1964), Cohen and Cohen (1975), Duncan (1975) and Li (1977). The methodological problems related to the study are discussed in Lindqvist (1980).

Design of the study

In the first study data were obtained by personal interviews. This was then followed by two waves of interviews conducted by telephone with approximately 200 and 500 randomly selected households.

Within households, the member in charge of the family economy, *i.e.* the head of the family, was interviewed. The household was made aware of this by an initial explanatory letter and by the interviewer before the formal interview was carried out.

An interview schedule with both open end and structured questions was used. Data were obtained on "declared" saving and, in addition to demographic data on education, income, family size and composition, employment and residence, subjective factors, like attitudes and expectations, which were assumed to cause changes in household's saving in the short- as well as in the long-run.

The interviews were designed for use on a regular basis—at three—or six months intervals.

The sample

Of the 548 households originally selected, 429 interviews were carried out in the spring of 1978. 45 households (including single households) were excluded from the sample because the respondents were dead or the names and addresses could not be verified, 20 were for various reasons unavailable throughout the period (the interviewer attempted to contact the household twelve times or more), and 54 refused to participate. The response and non-response rates are summarized in table 1.

With the exception of a slightly higher proportion of families with children in the sample, there were no differences in the household's

Table 1
Description of the sample.

	No. of households	Percent
Original sample size	548	
Excluded (dead, unknown)	45	
Sample size	503	100
Refusals	54	11
Noncontacts	20	4
No. of interviews	429	85

socioeconomic and income characteristics compared to the representative national samples of the National Central Bureau of Statistics.

Variables in the study

Dependent variables

Four dependent saving measures were used, three flow measures (changes in bank savings, savings in stocks and bonds, repayments of debts) and one stock measure (the amount the household could withdraw from the bank).

Saving was measured by the change in declared assets over a three-month period. Each respondent was asked about the household's possession of different types of banking accounts, stocks and bonds. For each type of saving the household was asked how much the balances had increased or decreased during the last three months. This is no doubt a rough estimate, but comparisons with the true values (bank-books, bank statistics) showed that the households usually were well aware of their balances.

Possession of and changes in credit (bank loans and installment payments) were examined in the same manner.

By summing up the changes in the various forms of saving, different saving measurements were created for each household. This is illustrated in table 2.

By limiting the questions to cover a certain time period, a key-problem was largely avoided. The private economy is a sensitive subject

Table 2
Different measurements of changes in wealth.

Types of savings		Changes during the last three months	
<i>Types of bank accounts</i> <i>Savings accounts</i> (incl the postal saving system): — Savings account with a special "premium" ^{a)} — Savings account with special loan possibilities ^{a)} — Deposit account, 6 months' notice or more — General savings account — Other accounts <i>Current accounts</i> : — Cheque salary account — Other current accounts	Bank savings (total)	Total savings excluding changes in debts (discretionary saving)	Total savings
	Other savings (total)		
Stocks			
Bonds			
Bank loans, mortgage loans	Total changes	(contractual	
Installment payments	in debts	saving)	

^{a)} These accounts are forms of contractual bank saving, where a fixed amount is withdrawn from the wages every month.

to study and many persons feel embarrassed when asked to disclose how much (or little) they have invested in the bank, in stocks, *etc.* This problem was largely avoided by limiting the questions to the more neutral *changes* over a certain time period. Admittedly, one could not get a clear picture of the size of the assets, *the stock*, of the various households. However, an indication of the *liquidity* was obtained through a question whether the household was able to withdraw a certain amount of money from the bank at short notice. The respondent was asked to state whether the household could withdraw specified amounts of money.

To summarize, the dependent variables were:

- bank savings during the last three months;
- changes in other securities (stocks and bonds) during the last three months;
- repayment of debts during the last three months;
- the amount the household could withdraw from the bank.

Independent variables

Socioeconomic characteristics

A large number of variables may be used to explain household's saving behavior. Socioeconomic characteristics, such as income and stage in life cycle, are all potential predictors of behavior. Saving attitudes and other economy-related attitudes may also guide household behavior and thus serve as predictors.

Only a few *a priori* explanatory variables could be included in the interviews. First of all the *socioeconomic characteristics* were included, such as *household income*, *stage in family life cycle*, *educational level* of husband and wife, *type of residence*. According to traditional micro-economic theory saving is influenced by socioeconomic variables. Chiefly income, change in income or stage in the family life cycle are used as predictors (for a review see Ferber 1971).

Obviously the type of dwelling (rented or owned) strongly affects the finances and the choice between discretionary and contractual saving. The type of dwelling chosen may be explained by the attitudes toward credit use, earnings, taxation rules and other factors with economic consequences such as the number of children.

The education level may also be expected to influence the choice between different types of saving. Lundman (1969), for example, showed that the education level was one of the most important variables to explain stock-ownership.

In addition to the socioeconomic characteristics, an attempt was made to indicate the household's *attitudes*, *expectations*, *economic activities* and *economic satisfaction*.

Attitudes and expectations

Only a small number of studies are aimed at the influence of psychological variables on saving and related economic activities. Attitudes towards saving are assumed to be deeply rooted and connected with upbringing and life style (Ölander and Seipel 1970; Julander 1975). It is therefore less likely that fluctuations will occur in the short run. Registered attitude changes toward saving and other economic activities during one period may, if attitude changes precede behavioral changes, serve as an indicator of long-term behavioral changes.

Attitudes passing by more rapidly may also influence the economic behavior. Many studies have shown that questions referring to expecta-

tions and plans may be useful for the measuring of short-term variations of economic behavior and be of predictive value for the short term (*cf.* Katona 1964; Katona *et al.* 1971; Strümpel *et al.* 1970; Berglund and Olsson 1971). The importance of expectations is also emphasized in studies using more complex aggregate models (*cf.* Tobin 1975; Juster and Taylor 1975; Markowski and Palmer 1977).

In the analysis of the saving-related attitude questions, an index consisting of 14 items was constructed. The internal consistency coefficient, using the McKennell formula for α (McKennell 1970), was 0.96.

With a view to measure expectations, questions similar to those used by Katona in his Consumer Sentiment Index were asked in the interviews. The household's optimism or pessimism was measured by three questions: whether they felt that their standard of living had increased or decreased during the last six months, the anticipated development during the next six months and whether they expected an increase or decrease in saving during the next six months. These questions formed an index with the consistency coefficient of 0.46.

Economic activity

The number of bank accounts and other types of saving, whether the respondent discussed saving and tried to protect himself against inflation were assumed to express an economic interest and active planning of the finances. The economic activity was expected to influence saving behavior in the same way as attitudes. After standardizing these variables they were put together in an index with the rather low consistency of 0.40.

Economic satisfaction

The subjective satisfaction ratings were assumed to be a consequence of the economic situation and the economic behavior of the household. These data were based on questions of the household's content or discontent with the income and questions whether it was easy or not to make the income last out. The consistency coefficient of the index was 0.57.

Such measures, that have the individual's or the household's subjective well-being in view, represent important social indicators (*cf.* Strümpel 1976; Fox 1974).

To summarize, the independent variables or groups of variables were:

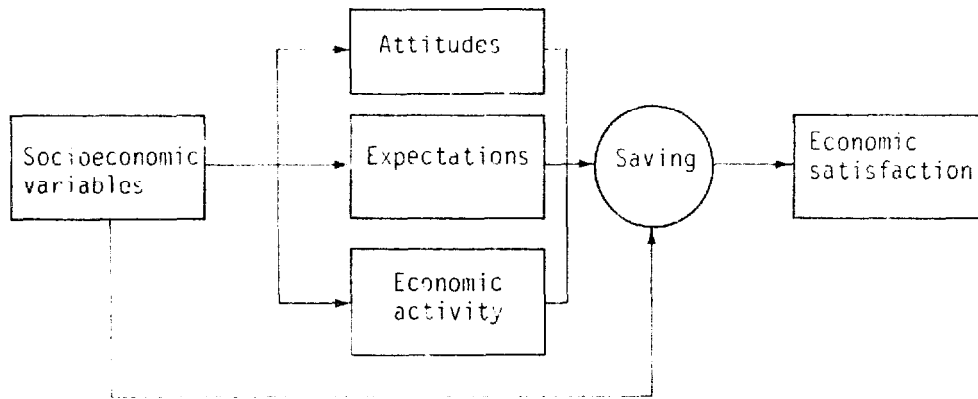


Fig. 1.

- socioeconomic variables;
- attitudes;
- expectations;
- economic activity;
- economic satisfaction.

Interrelationships

The hypothetical interrelationships of the criterion variables and the predictor variables/groups of variables are given in a simple causal model in fig. 1.

Explaining saving behavior

According to fig. 1 the saving behavior is explained by the socio-economic characteristics of the households and by “softer” variables such as saving-related attitudes and interests. In the following, these variables’ effects on the different aspects of household saving are analyzed separately using stepwise multiple regressions and with regard to their hypothetical causal order using path-analysis.

Results of the regression analysis

To test the influence of each independent variable upon saving, a number of regressions were performed, using three of the saving mea-

surements in table 2 and the liquidity-estimate as dependent variables.

Table 3 shows the results of the regression analyses. The values given in table 3 are partial regression coefficients (beta-weights) and may be interpreted as the influence of each independent variable upon the set of saving measurements (with adjustments for all other independent variables).

The overall accuracy of the predictor equation is reflected by R^2 , the proportion of variation explained by all the variables included in each regression equation. However, what is more interesting than the overall explanatory power of the model is the relative importance of the different predictors in accounting for the variance in the dependent variable.

Bank savings

None of the included variables were found to contribute substantially to explaining bank savings during the three-month period. Only the satisfaction index—treated as an effect variable in fig. 1 but here as a predictor—showed a weak relationship with the amount saved in banks. That is: the higher the bank savings during the period, the more pleased the respondents were with the economic situation.

A more surprising result was that none of the socioeconomic variables, and the in economic theory important variables, were related to bank savings. It can be assumed that variables like household income should be important predictors, but in the micro level analysis they showed no predictive value.

Why then cannot bank savings be explained? One possible reason is the selection and measuring of the predictors. Another is the definition and measurement of bank savings [1]. Other analyses, not reported here, show that different saving objectives are considered important by the households. Briefly, it may be maintained that bank saving is often an activity which either reduces the feelings of financial uncertainty or is directly aimed at a more or less distant consumption goal. These objectives vary among the households and can compensate for one other. This is, however, not reflected by the measure used as dependent

[1] Here it is defined as the change over a three-month period, which is a short-term measure. Measurements of the regularity of saving behavior—together with the liquidity estimate reported in this paper—may be considered as more long term measurements.

Table 3

Standardized regression coefficients for variables predicting household saving. The order of inclusion is put in brackets.

Predictors	<i>Measurements of changes in wealth (dependent)</i>			
	Bank savings	Repayments of debts	Total savings	Liquidity (Bank funds)
1. Income	0.07	0.18 ^{b)} (3)	0.14 ^{a)} (1)	0.04
2. Stage in family life cycle	-0.05	-0.09	-0.11 ^{a)}	0.09
3. Family size	0.02	0.16 ^{b)} (1)	0.08	-
4. Type of residence	0.04	0.22 ^{a)} (2)	0.12 ^{a)} (2)	0.07
5. Education	-	-	0.03	0.07
6. Attitudes	0.10	0.06	0.13 ^{a)} (3)	0.07
7. Expectations	-0.04	-	-0.03	0.09 ^{a)} (3)
8. Economic activity	0.08	-0.03	0.06	0.26 ^{b)} (1)
9. Economic satisfaction	0.11 ^{a)} (1)	-0.14 ^{b)}	0.03	0.24 ^{b)} (2)
Multiple R	0.22	0.44	0.33	0.46
R ²	0.05	0.19 ^{b)}	0.11 ^{a)}	0.21 ^{b)}

^{a)} significance at $p < 0.05$

^{b)} significance at $p < 0.01$

Note:

1. High family income high score, low income low score
2. Six categories ascending
3. Ascending
4. Rent = 1, own = 2
5. High educational level high score, low educational level low score
6. Index score. Positive attitudes high score, negative low score
7. Index score. Positive expectations high score, negative low score
8. Index score. High activity high score, low activity low score
9. Index score. Low satisfaction low score, high satisfaction high score. This variable is treated as an effect of saving in fig. 1, i.e. it is a dependent variable. However, in the regressions reported in table 3, the variable is introduced parallel with the predictors.

in the regression. If the bank savings during the period could be decomposed into sums attributable to each objective, this might increase the variance explained by the predictors used in the equations.

Repayments of debts

In contrast to bank savings, the measure of household income was contributing to the variation in changes in debts. High-income house-

holds were expected to get long-range credits more often and they were also expected to be more active in demanding credits—because of the income taxation rules among other things. This was not, however, confirmed by the activity index, but further analyses showed that the activity index had predictive value for a distinct sub-group consisting high-income (and high-status) households. The low beta-weight was partly an effect of multicollinearity.

The type of residence and the family size also contributed substantially to explaining debts. The dependence of debts on type of residence is obvious. The residence type is, however, a prerequisite for getting new credits over and above the mortgages, which justifies the inclusion of the variable also if short-term measurements are made regularly.

The relationship between debts and family size was found to be a relationship between family size and mortgages (and not more short-run credits) and shows how the number of children affects the claims for housing standard.

Psychological measurements, such as attitudes and expectations, had no explanatory value. Only the satisfaction index was significantly related to debts. The sign of the coefficient indicates the direction of the relationship which was the opposite compared to bank savings. Dissatisfaction was more frequent among the households with larger possessions of and changes of credits (bank loans and installment payments).

Total savings

The change in total savings is an overall measurement including not only bank savings and debts but also changes in holdings of stocks and bonds.

This is not a “natural” saving measure at the level of individual households in the sense that they do not think about their savings in such a way. But from the point of view of society it is a relevant estimate since total savings of households express what is not intended for consumption during the defined time period and can be aggregated to give a meaningful indicator at the macro level.

The most important predictor was the family income. Of importance were also the family life cycle and the closely related type of residence. The pattern of effects of these variables was known from the debt measure.

The positive effect of saving-related attitudes on total savings indicated a higher valuation of saving among the households with the highest saving scores.

Liquidity

The fourth dependent variable in table 1 was an indication of the liquidity, and was not computed in the same way as the change measurements. The respondents were asked whether the household could withdraw specified amounts of money up to 10000 crowns (or more) from the bank.

The estimates of bank savings, debts and total savings used here measure the change during a relatively short time period (three months). The bank liquidity, on the other hand, is mostly a sign of more long-term savings.

As in the other regressions, only a few included variables contributed to the variation in the dependent variable. In this equation the intervening activity index and expectations were related to liquidity together with the satisfaction index. None of the socioeconomic variables were directly related to the estimate of the dependent variable.

The activity index turned out to be the most important predictor. The index, which was intended as an indication of economic planning and interest, could alone explain 12 per cent of the variation in bank liquidity. The sign of the coefficient shows that the highest liquidity, the largest bank funds, were found among the more active households.

The relationship appeared at first contradictory since households interested in protecting themselves against inflation (one of the components in the index) ought to use more inflation-proof investments than bank accounts. But for all types of households, bank savings seemed to be the primary concern because of its high liquidity, as buffer for security or emergencies and since it was a prerequisite for other activities.

This uncertainty aspect was also indicated by the dependence of bank liquidity on economic satisfaction and expectations. People with larger bank savings seemed to be more satisfied and optimistic about their economic situation.

A path-analytic approach

In the regression equations the direct effects of the predictors upon the dependent variables were illustrated. With regard to the aims of the project, it is important to study the direct effects, but also to obtain knowledge of the underlying relationships. This usually requires time series data, but by introducing hypothetical relationships among the predictors (*cf.* fig. 1) into the analysis, the obtained cross-sectional regression results can be decomposed in terms of direct and indirect relationships.

This estimation procedure, path analysis, is similar to that of multiple regression analysis, using ordinary least squares estimation to express the relationships between the variables, but the estimation requires a more detailed causal model than given in fig. 1. The requirements and assumptions for path analysis are discussed by Duncan (1966, 1975), Asher (1976) and Li (1977).

A postulated causal model for one of the saving estimates

In analogy with the simple S-O-R-model in fig. 1 and with earlier theoretical and empirical research a more detailed causal model was constructed for analysis of the household's bank liquidity—the last dependent variable in table 1 and the one with the highest R^2 value.

The socioeconomic characteristics were assumed to influence bank liquidity directly and indirectly via the intervening variables. Family size, stage in family life cycle and educational level were considered as exogenous variables causally related to household income and type of residence, in turn considered to determine variables expressing attitudes, expectations and economic activity, which were assumed to influence behavior directly.

Variables for which the path coefficients did not show a significant value ($p < 0.05$) were considered as having no effect on the dependent variable and were removed from the model. The somewhat less complex model and the values of the path coefficients are given in fig. 2. (Dotted arrows indicate the influence of residual variables.)

The total association between an independent and a dependent variable (measured by the correlation coefficient) can now be decomposed. The path coefficients represent the *direct* effect of the independent variable, and the causal structure in the path diagram implies that

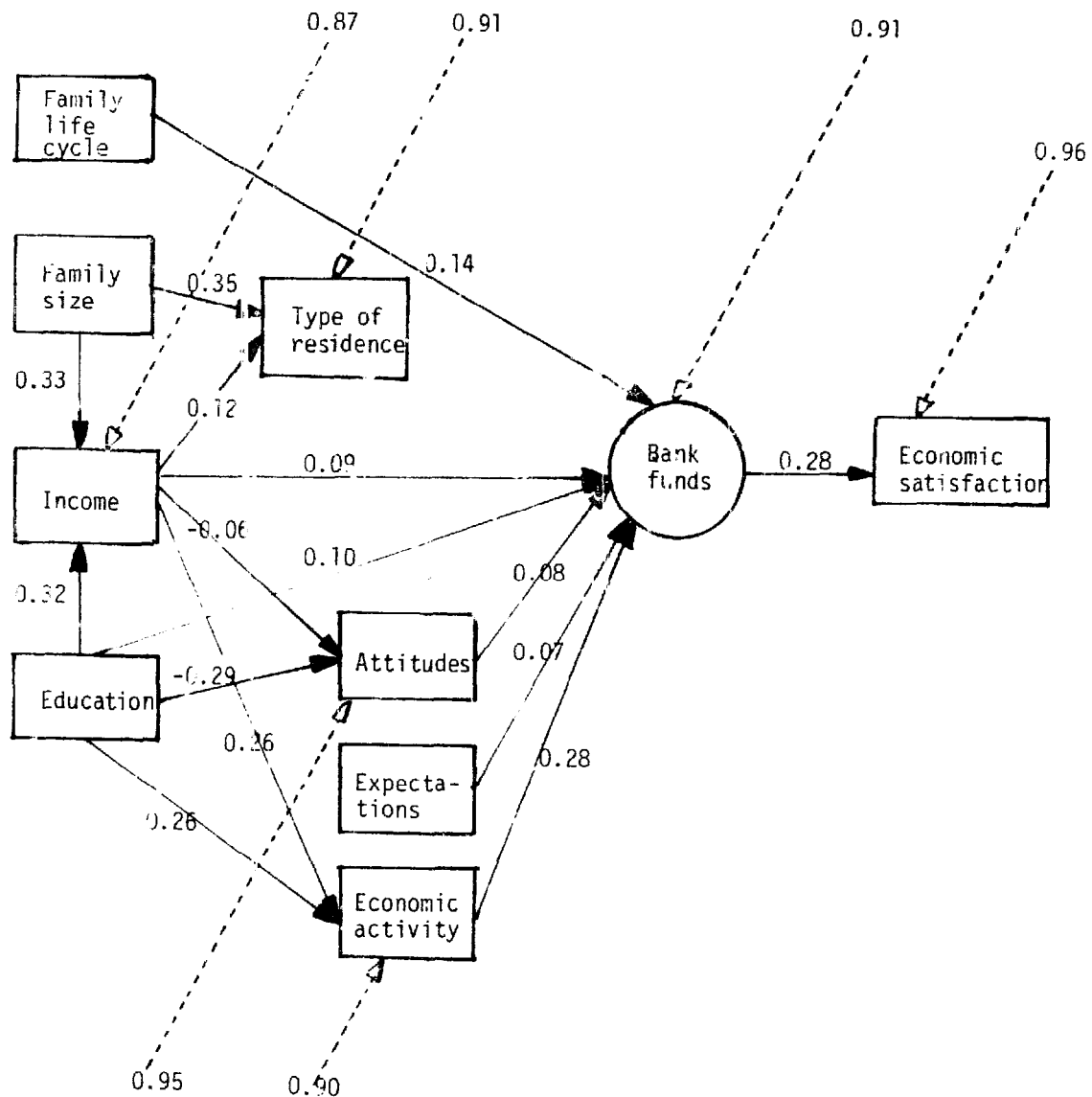


Fig. 2. Path diagram showing the observed path coefficients.

the path coefficients can be given a causal interpretation. The remaining components in the total association thus represent *indirect* causal effects (the effect of an independent variable is mediated by an intervening variable) and spurious effects (due to common causes or correlation between predetermined variables).

The decomposition of the total association between bank liquidity and the predictors in fig. 2 are given in table 4.

The ability of postulated sets of relationships in explaining the observed correlation is shown by the residuals in the right column in table 4. Usually the observed correlations could be satisfactorily reproduced by the model, but one exception was the expectations where only approximately half of the observed correlation with bank liquidity

Table 4
Decomposition of the total association between bank liquidity and the predictors in fig. 2.

Dependent variable	Independent	Observed correlation A	Due to the model			Unexplained by the model $A - (D + E)$
			Direct effect B	Indirect effect C	Causal effect $D = B + C$	Non-causal E (spurious)
Bank funds	Family cycle	0.295	0.1420	—	0.1420	0.0145
	Income	0.1923	0.0947	0.0781	0.1728	0.0330
	Education	0.1678	0.1011	0.1036	0.2047	0.0425
	Attitudes	0.1652	0.0756	—	0.0756	0.0219
	Expectations	0.0805	0.0679	—	0.0679	0.0199
	Ec. activity	0.3407	0.2792	—	0.2792	0.0550
						0.0270
						0.0135
						0.0056
						0.0115
						0.0325
						0.0065

could be explained by the model. The effect of expectations, and attitudes, was however not very strong.

The most significant effect on the household's bank liquidity was caused by the activity variable. Both the direct and the total causal effect showed the highest scores (total = 0.28). Also the causal effect of the household's educational level, which directly and indirectly was expected to influence saving via the intervening variables, was substantial (total = 0.20).

Despite the regression results known earlier, the household income had a considerable effect on the liquidity estimate. Like the education level, the indirect influence via the intervening variables were substantial.

The decomposition of the association into direct, indirect, and spurious components illustrate how the path analysis can be used as a complement to regression analysis.

Conclusions

The study has primarily dealt with the methodological issues—to develop instruments for regular use serving as indicators and predictors for household's saving at the micro level. In this paper the influences of some selected predictors on different types of household savings were examined using cross-sectional data.

Three flow measures, *i.e.* reported bank savings, repayments of debts and total savings (a summation of reported bank savings, savings in stocks and bonds and repayments of debts), and one stock measure, reported bank liquidity, were analyzed separately. Both socioeconomic and income characteristics and “softer” variables, such as attitudes and expectations, were used as explanatory variables.

In the regression analyses, the variables introduced failed to explain bank savings, while the traditional socioeconomic variables influenced the debt measure, and the softer ones were more successful in explaining the stock measure. When the latter regression result (with the highest R^2 value) was decomposed in direct and indirect effects using a hypothetical causal model and path analysis, it appeared that the effects of household income and educational level were substantial. However, the intervening variables were still significant.

It would be presumptuous to claim that the results obtained can be

generalized. The sample is small and time series data are required for an analysis of the causal interrelationships. Nevertheless, some conclusions may be drawn from the results of the study. In the first place, the explanatory value of the set of predictors is low. Only a few, and different, predictors are related to the four saving estimates used in the study. Secondly, both traditional economic and intervening variables seem to influence the household's saving behavior. This is further indicated by the path analysis. Thirdly, the results indicate that the different categories of saving ought to be studied separately when trying to explain household saving behavior since different phenomena may be involved.

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